

Project Design Phase

Solution Architecture

Date	19 July 2026
Team ID	[Enter Your Team ID]
Project Name	BookEase – Multi-Role Online Bookstore Platform
Maximum Marks	4 Marks

BookEase bridges the business need — a trustworthy multi-role bookstore — with a standard 3-tier MERN architecture, described below.

Solution Architecture Diagram

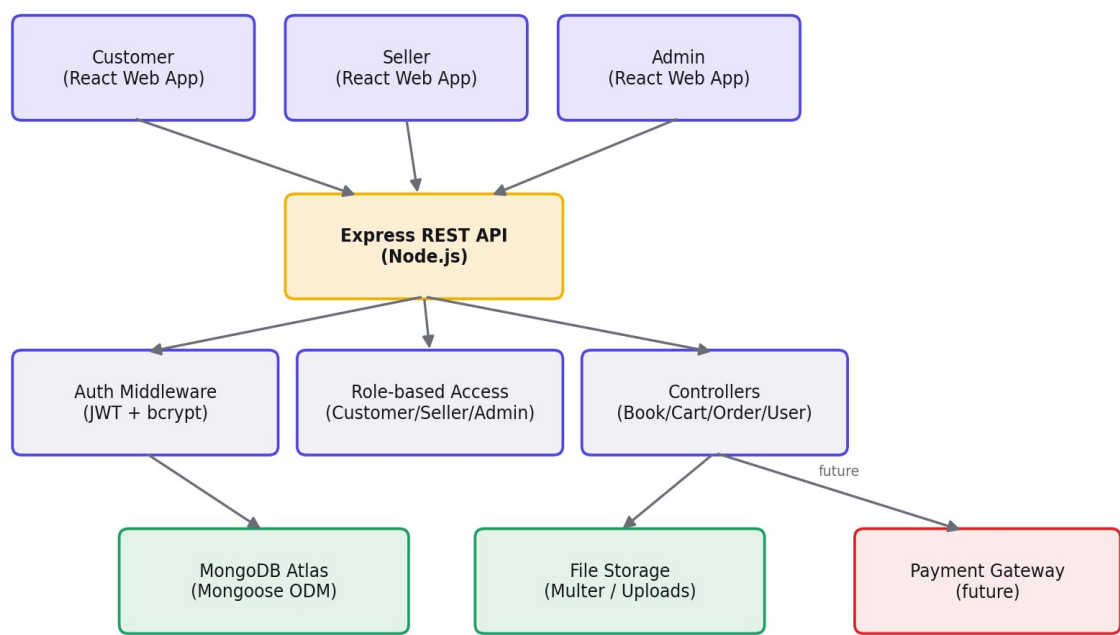


Figure 1: BookEase solution architecture — React clients for each role, a single Express REST API secured by JWT and role-based middleware, MongoDB Atlas for data, and Multer-based file storage for images.

Architecture Layers

Layer	Responsibility	Key Technology
Presentation	Role-specific UI for Customer, Seller and Admin; client-side routing and state (auth, cart)	React.js, React Router, Context API
API / Gateway	Single Express REST API exposing versioned endpoints for auth, books, cart, wishlist, orders, users	Node.js, Express.js

Layer	Responsibility	Key Technology
Security	Token issuance & verification, password hashing, role-based route guards	JWT (jsonwebtoken), bcrypt, custom middleware
Business Logic	Controllers implementing catalog, cart/checkout, order fulfillment, and admin workflows	Express controllers (bookController, cartController, orderController, userController)
Data	Persistent storage of users, books, categories, carts, orders, reviews and wishlists	MongoDB Atlas via Mongoose ODM
File Storage	Book cover images and profile pictures	Multer (local uploads); Cloudinary planned for production
External (planned)	Online payments and transactional email	Razorpay/Stripe; Nodemailer/SendGrid